

How Small is Small?

Non-linearities in Heterogeneous Agent Models ^{*}

Javier Bianchi[†] Greg Kaplan[‡]

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Abstract

In plausibly calibrated heterogeneous-agent models, marginal propensities to consume (MPCs) are highly non-linear in wealth, falling sharply away from borrowing constraints. As a result, the aggregate consumption response to a fiscal transfer is strongly concave in its size: larger transfers shift households out of high-MPC regions and thereby dampen the consumption response. Across partial- and general-equilibrium settings, linear methods substantially overstate the effects of fiscal stimulus at empirically relevant sizes. Local methods are not reliable for studying shocks and policies where a failure of Ricardian equivalence is important.

Keywords: Heterogeneous agents, non-linearities, HANK

JEL classification: D15, D31, D52, E21, E62, G51, C63

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[†]Federal Reserve Bank of Minneapolis, javier.i.bianchi@gmail.com.

[‡]University of Chicago, e61 Institute and NBER, gkaplan@uchicago.edu.

1 Introduction

There has been an explosion of work studying the effects of fiscal and monetary interventions in settings with heterogeneous agents (HA) and incomplete markets. These frameworks are useful because they do a good job of capturing relevant features of household-level spending and savings behavior. In particular, suitably calibrated versions can match both the level and distribution of marginal propensities to consume (MPC).

These aspects of consumption behavior are especially relevant for studying fiscal interventions, because a failure of Ricardian equivalence lies at the heart of the aggregate and distributional effects of such policies. The size and pattern of MPCs underscore these departures from Ricardian equivalence and drive much of the difference between heterogeneous-agent and representative-agent models. Indeed, the effects of fiscal stimulus programs funded by future taxes are driven entirely by such departures.

A large part of the recent literature has analyzed these settings using various forms of first-order methods.¹ The most common approach, labeled “sequence space Jacobian” (SSJ) by Auclert, Bardóczy, Rognlie and Straub (2021), computes an aggregate consumption function by aggregating consumption policy functions across households, and computes equilibria via a linear approximation of this function around its steady state in a suitably chosen vector of aggregate prices and policy variables. It is well understood that such methods approximate true equilibria well when the policy innovations and exogenous disturbances being studied are small. But how small is small?

We argue that for commonly studied fiscal interventions, small is indeed very small—much smaller than many of the aggregate shocks and interventions that macroeconomists are motivated to explore based on real-world experience.

Our starting point is the observation that for a funded fiscal stimulus program, the bulk of the aggregate effects are driven by the strength of departures from Ricardian equivalence, which in turn are driven by the extent to which households’ MPCs exceed those implied by the permanent income hypothesis (PIH). HA models can be calibrated to match quarterly MPCs observed in the data and thus are a useful class of models to study funded fiscal stimulus, unlike RA models for which such policies have no aggregate effects. However, the MPC functions in suitably calibrated HA models are highly non-linear in wealth: they decline sharply away from borrowing constraints and kinks. It follows that MPCs are highly

¹Examples include Aggarwal, Auclert, Rognlie and Straub (2023), Bardoczy, Sim and Tischbirek (2024), Summers and Rachel (2019), Hänsel (2024), Campos, Fernández-Villaverde, Nuño and Paz (2024), Eichenbaum, Guerreiro and Obradovic (2025); Angeletos, Lian and Wolf (2024b,a).

non-linear in the size of a transfer, as large transfers move households away from constraints and substantially lower their MPCs. This observation leads us to conjecture that inferring the aggregate effects of a fiscal intervention on the basis of the aggregate consumption response to very small disturbances may be misleading. In the remainder of the paper, we investigate when this is and is not the case.

Overall, our findings do not bode well for using first-order methods, such as SSJ, to study the effects of fiscal interventions. Moreover, the features of household consumption that are at the heart of the issue suggest that even higher-order local methods are unreliable for studying this class of policies. The reason is that the consumption function for constrained households, whose behavior is quantitatively relevant for the effects of these policies, exhibits a kink at the level of wealth where the constraint stops binding.

We start in Section 2 with a partial equilibrium exercise that highlights the key mechanism behind the non-linearity and illustrates its quantitative significance. In Section 3, we then analyze various general equilibrium models with different assumptions about production, nominal rigidities, fiscal financing and monetary policy. In Section 4 we provide examples of shocks and other policy interventions where linear solutions approximate non-linear solutions much better. Our simulations suggest that the non-linearities we emphasize are significant when a failure of Ricardian equivalence is an important component of the transmission mechanism. Consequently, those shocks and policies for which local methods are accurate in HA models tend to be those where aggregate responses are similar in RA and HA economies.

2 Partial Equilibrium

2.1 Model

Demographics Time is discrete and infinite $t = 0, 1, \dots$. The economy is populated by a measure-one continuum of households that face uninsurable idiosyncratic shocks. We assume the path for aggregate disturbances is known at $t = 0$, so there is no aggregate uncertainty.

Preferences Households have utility over consumption, where $u(c_{it})$ is strictly increasing and concave, and discount the future with factor β_i , which is heterogeneous across households.

Income and assets Each period households receive earnings y_{it} whose realizations are independent across households. We describe the stochastic process for earnings below.

Households pay taxes according to a tax-and-transfer schedule $\mathcal{T}_t(y)$ that depends on their earnings. A positive value of $\mathcal{T}_t(y)$ is a payment to the government, while a negative value is a transfer from the government. They can save and borrow in a one-period risk-free real asset which pays interest rate r .

Household problem Households enter period $t - 1$ with assets $a_{i,t-1}$ and choose next-period assets $a_{i,t}$. They face the borrowing constraint $a_{i,t} \geq \underline{a}$. The household problem is

$$\max_{\{c_{i,t}, a_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_i^t u(c_{i,t})$$

subject to

$$c_{i,t} + a_{i,t} \leq (1 + r)a_{i,t-1} + y_{i,t} - \mathcal{T}_t(y_{i,t}) \quad \text{and} \quad a_{i,t} \geq \underline{a},$$

The first-order condition of this problem yields the intertemporal Euler equation:

$$u'(c_{i,t}) \geq \beta_i(1 + r)\mathbb{E}_t u'(c_{i,t+1})$$

with equality if $a_{i,t} > \underline{a}$.

Aggregation The solution to the recursive version of the household problem yields time-varying decision rules for consumption $c_t(a, y, \beta)$ and next period's wealth $a'_{t+1}(a, y, \beta)$. Given an initial distribution over assets and income and discount factors, which we denote by $\Gamma_0(a, y, \beta)$ and a path for the tax-and-transfer schedule $\mathcal{T} = \{\mathcal{T}_s\}_{s=0}^{\infty}$, these decision rules induce a path of distributions $\{\Gamma_t(a, y, \beta)\}_{t=0}^{\infty}$, which implies a path of an aggregate consumption function that gives total consumption in period t , which we denote by $\mathcal{C}_t(\mathcal{T})$.

2.2 Steady-state calibration

Demographics A period represents a quarter. The numeraire in the model is quarterly GDP per household which was around $\$166,000/4 = \$41,500$ in 2019.

Income and assets We set the interest rate to 2% p.a. and the borrowing limit to $\underline{a} = 0$. We model the process for log earnings $\log y_{it}$ as the sum of two orthogonal components, an AR(1) component and an IID component. Following [Kaplan and Violante \(2022\)](#), we assume that shocks to both components arrive stochastically with a Poisson arrival rate of 1/4, so that shocks are received on average once a year. We set the parameters of the earnings

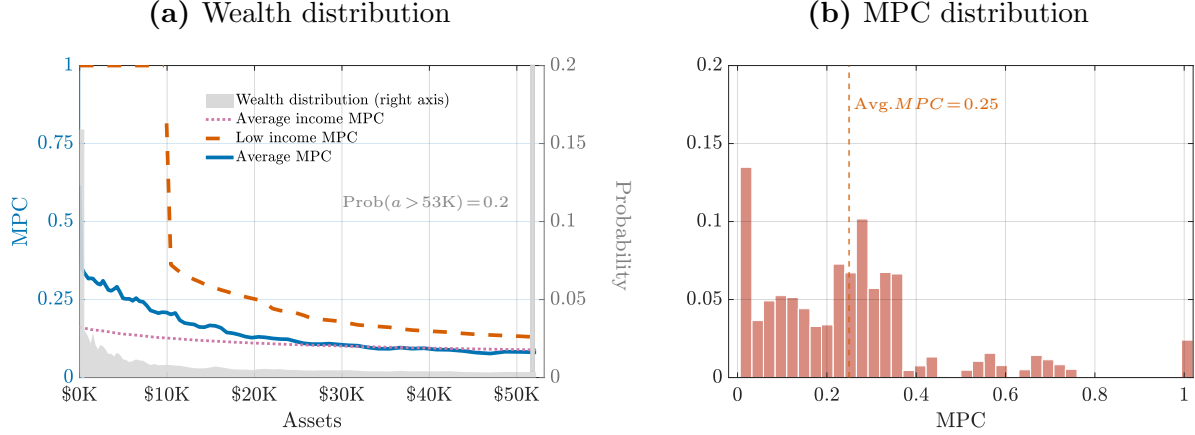


Figure 1: Steady-state wealth distribution and MPCs.

Note: The left axis of panel (a) presents the MPC for average-income and low-income households and the average MPC across households, both as a function of assets. The right axis of panel (a) presents the wealth distribution up to \$50K. Panel (b) presents the distribution of MPCs.

process to match moments of the household labor income distribution from the Panel Study of Income Dynamics. See Appendix A for details.

Preferences We assume $u(c) = \log(c)$. We assume the discount factor β_i takes seven equally spaced values. We choose the mean discount factor so that the ratio of mean wealth to annual GDP is 100% (\$166,000). We choose the dispersion in the discount factors so that the average quarterly MPC out of a one-time \$1,000 windfall is 25%. The implied range of discount factors is between 0.947 and 0.993.

Tax and transfers The tax system consists of a proportional tax rate and lump-sum transfer, whose values in steady state we denote by τ and T , respectively. Net earnings are given by $y_{it} - \mathcal{T}(y_{it}) = (1 - \tau)y_{it} + T$. We set $\tau = 0.274$ to match total government revenue of 27.4% of GDP in 2019. We set the steady-state lump-sum transfer $T = 15\%$ of GDP, which is around \$25,000 per household per year.

Wealth and MPC distributions Figure 1a shows the stationary wealth distribution overlaid with the MPC as a function of assets, for households with different income levels.² Figure 1b shows a histogram of these MPCs. Two features of MPCs in this class of models

²We plot the MPC out of an infinitesimal windfall because this is the MPC that is relevant for the linear vs non-linear comparisons that are the focus of this paper. We calibrate to the MPC out of a \$1,000 windfall because that is more consistent with the empirical evidence we match to.

that drive our aggregate findings are apparent. First, the average MPC is declining in wealth and approaches the PIH MPC as assets rise. This decline reflects two forces: (i) at each income level the MPC is weakly declining in wealth; and (ii) very low-wealth households are disproportionately those with low income, who are borrowing constrained at low asset levels and have an MPC of 100%. As wealth increases, the mix shifts towards high-income households who have much lower MPCs.

Second, the distribution of MPCs is highly dispersed and approximately bi-modal. A subset of households with both low assets and low income have very high MPCs, close to 100%. For these households, the MPC function is flat at 100% in a small region above the borrowing constraint, after which it drops discontinuously. The next section explores the implications of these patterns for the shape of the aggregate consumption function.

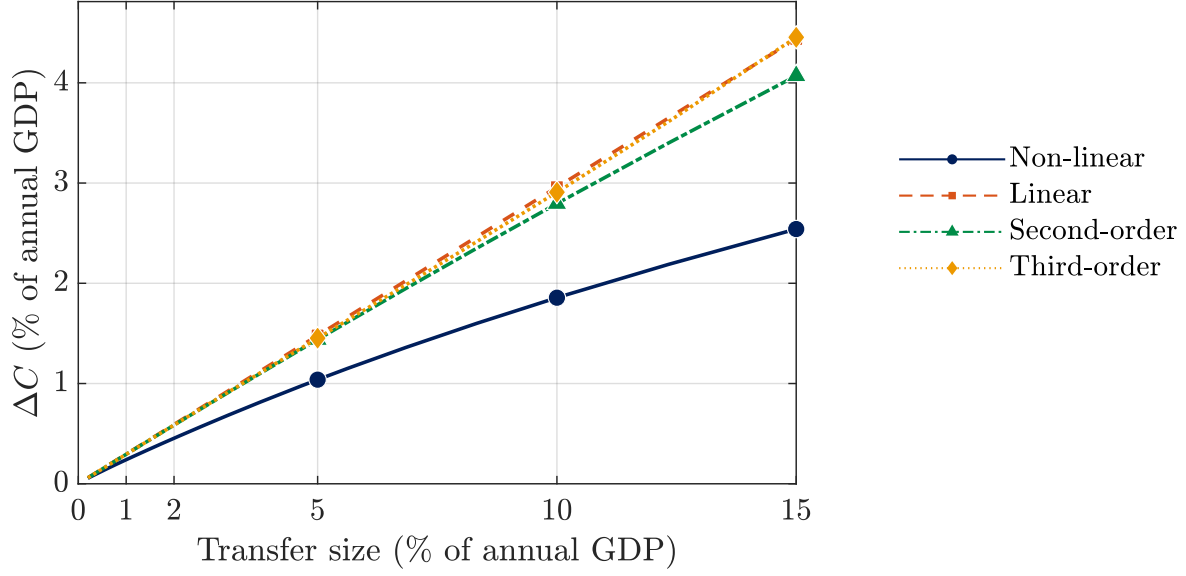
2.3 Aggregate Transfers in Partial Equilibrium

Experiment Starting from the steady-state distribution, we study an unexpected one-time increase in the lump-sum transfer to all households at $t = 0$. We implement this by setting the transfer to $T_0 = T + \Delta$ and keeping $T_t = T$ for $t > 0$. Since the only change in the tax-and-transfer schedule is at $t = 0$, we denote aggregate consumption at time t by $\mathcal{C}_t(\Delta)$. Our interest is the aggregate consumption response, $\mathcal{C}_t(\Delta) - \mathcal{C}_t(0)$.

Computation We compute steady-state decision rules using the method of endogenous grid points. Since we are studying a one-time transfer at $t = 0$ in partial equilibrium, we can compute the non-linear response to the transfer shock using the steady-state decision rules and propagating forward the household wealth distribution.

We compute the linear response using a finite difference in Δ to approximate the derivative of the aggregate consumption function $\frac{\partial \mathcal{C}_t(\Delta)}{\partial \Delta}$. In partial equilibrium this is equivalent to how the transfer is treated in the SSJ method that we use for general equilibrium experiments in Section 3. We have also verified that computing $\frac{\partial \mathcal{C}_t(\Delta)}{\partial \Delta}$ using automatic differentiation yields the same result as our central finite-difference approximation, as well as the analytical derivative (B.13) in Appendix B, with differences of order 10^{-7} . See Appendix C for details of computation.

(a) Aggregate effects of a one-time transfer in partial equilibrium



(b) Linear to non-linear aggregate consumption response

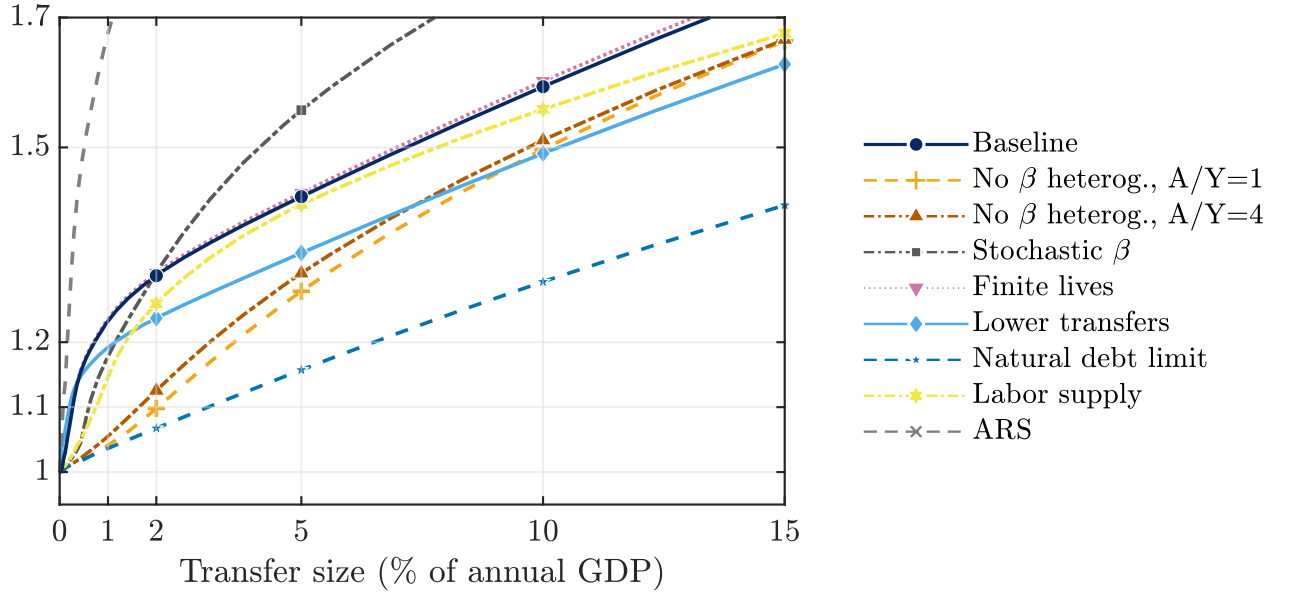


Figure 2: Partial equilibrium effects of a one-time transfer.

Notes: Panel (a): ΔC is expressed as a percentage of annual GDP. Panel (b) presents the ratio of the linear to the non-linear consumption response for different model calibrations. See Table A.3 in Appendix A.2 for more details.

Linear vs. non-linear responses Figure 2 contains the main results of the partial equilibrium exercise. Figure 2a shows the first-quarter increase in aggregate consumption $\mathcal{C}_0(\Delta) - \mathcal{C}_0(0)$, expressed as a percentage of annual GDP per household, as a function of the size of the transfer Δ . The blue solid line labeled “Non-Linear” shows the results computed using the non-linear solution and the red dashed line labeled “Linear” shows the results from the linear solution. The aggregate consumption response is concave in the size of the transfer. For very small transfers, the linear and non-linear solutions produce similar aggregate consumption responses. But for larger transfers, the linear solution substantially overstates the consumption response. For example, for a transfer that is 5% of annual household GDP (around \$8,400), the linear solution overstates the non-linear solution by around 42%. Even for a transfer of 0.5% of annual GDP (around \$840), the linear solution overstates the non-linear solution by more than 17%.

Alternative calibrations Figure 2b shows that the concavity of $\mathcal{C}_0(\Delta)$ is a robust feature of this class of models and is not specific to our particular calibration and choice of model features. The figure shows the ratio of the linear consumption response to the non-linear response for transfers of different sizes, for a range of alternative calibrations and versions of the model. We report the results in terms of the ratio of the linear to non-linear solutions because the size of the response differs across models.

The versions of the model included in the figure include those without discount factor heterogeneity, with stochastic discount factors, with different levels of aggregate wealth, a life-cycle version with finite lives, with lower lump-sum transfers, with a natural borrowing limit, with endogenous labor supply, and with the idiosyncratic earnings process from Auclert et al. (2024). In all these versions, and many others we have experimented with, the linear solution overstates the aggregate consumption response by an economically meaningful magnitude for transfer sizes larger than 2% of GDP. In many cases, including our baseline, the errors appear at much smaller transfer sizes.

Intuition for concavity Ultimately, the challenge for local approximations of $\mathcal{C}_0(\Delta)$ around $\Delta = 0$ is not merely that the aggregate consumption function is strongly concave. It is that behind this concavity is a subset of households—those who are at the borrowing constraint—whose consumption functions are exactly linear in a small region around the constraint. The slope of this linear segment is 1, reflecting that for small windfalls these households have an MPC of 100%. Above this linear segment their MPC drops discontinuously. This discontinuity

can be seen in the red dashed line in Figure 1.³ In Appendix B we provide a proof that the derivative of the consumption function is discontinuous at the boundary of the linear segment. Figure E.4 displays how larger transfers reduce the share of households that remain constrained at $t = 0$. We also report additional figures to disentangle the role of the kink, versus the curvature of the consumption function for unconstrained households, in driving the non-linearity of the aggregate consumption function. These constrained households account for a large portion of the difference between the linear and non-linear responses. They also account for the bulk of the difference between aggregate consumption dynamics in HA models versus corresponding RA models.

Higher-order approximation The difference between the linear and non-linear aggregate consumption responses stems from the concavity of $\mathcal{C}_0(\Delta)$. It is natural to conjecture that a more accurate solution could be obtained by taking a higher order approximation around $\Delta = 0$. However, Figure 2a shows that second- and third-order approximations to $\mathcal{C}_0(\Delta)$ in fact offer little improvement over the linear approximation. To interpret these higher order approximations there are a number of subtleties to address.

First, note that there are two sources of non-linearity in $\mathcal{C}_0(\Delta)$: (i) concavity of the consumption function for unconstrained households; (ii) the kink in the consumption function for constrained households. As discussed above, the latter source of non-linearity only kicks in at a strictly positive $\Delta > 0$. Below this size of transfer, the consumption function is linear. Hence for constrained households the higher order corrections to the consumption function around $\Delta = 0$ are exactly zero.

Second, the endogenous grid point solution method that we use to solve for optimal consumption policies is in fact locally linear because we use linear interpolation between grid points. So even for unconstrained households, automatic differentiation (correctly) yields exactly zero for higher order terms in the approximation. One could use something other than linear or bilinear interpolation to interpolate between grid points, such as cubic splines or piecewise cubic Hermite polynomials. But the resulting higher order terms in the approximation of $\mathcal{C}_0(\Delta)$ would be merely a reflection of whatever assumed non-linear interpolation technique is chosen.

So to construct an accurate higher order correction to the consumption function for unconstrained households, we use the value of consumption at each level of assets between grid points that is implied by the Euler equation, and construct the implied higher order derivatives by using implicit differentiation. See Figure E.1 for details. These give rise to the

³See Figure E.3 in the Appendix E for the corresponding consumption function.

second- and third-order consumption responses reported in Figure 2a.

The reason why the quality of these highly accurate local approximations around $\mathcal{C}_0(\Delta)$ deteriorates rapidly for higher values of Δ is that it is the kink in the consumption function for constrained households, rather than the concavity for unconstrained agents that drives the bulk of the non-linearity. This is why even with exact computation of local derivatives, the radius of convergence for local approximations of any order is likely to be small. For constrained households, consumption behavior in the neighborhood of $\Delta = 0$ is simply not informative about consumption behavior for larger values of Δ because of the discontinuity in the MPC.

Yet another approach is to try to account for the kink by using finite differences in Δ with a larger step size. In Figure E.1 we report a wide range of examples that use Richardson extrapolation to combine approximations at different step sizes to cancel leading error terms. The figures show that finite difference approximations are extremely sensitive to the choice of step size and are typically very poorly behaved even for moderate shocks. Without knowledge of the non-linear solution it is unclear how one could use higher order finite difference approximations around $\Delta = 0$ to obtain reliable solutions.

3 General Equilibrium Model

In this section, we study the extent to which the non-linearity of the aggregate consumption function impacts the general equilibrium effects of one-time fiscal stimulus programs. We extend the model from the previous section to include production, labor supply, asset market clearing and nominal rigidities, as in canonical heterogeneous agent New Keynesian models (e.g., Kaplan and Violante, 2014; Auclert et al., 2023).

3.1 Model

Preferences Households have utility over real consumption c_{it} and hours worked n_{it}

$$\mathbb{E}_0 \sum_{t \geq 0} \beta_i^t [u(c_{it}) - v(n_{it})]$$

where $v(\cdot)$ is a strictly increasing convex function in hours worked.

Income and assets Real household income consists of earnings $e_{it} \frac{W_t}{P_t} n_{it}$ and dividend income d_{it} , where e_{it} is an idiosyncratic productivity shock that follows a process analogous to the one for y_{it} in Section 2. In this section we denote total non-asset income as $y_{it} = e_{it} \frac{W_t}{P_t} n_{it} + d_{it}$. With this notation, the borrowing constraint and budget constraint are the same as in Section 2, with r replaced by r_t .

Labor supply In our baseline model we assume flexible prices and sticky wages. The aggregate effective labor input $N_t = \int_i n_{it} e_{it} di$ is chosen by a labor union and is allocated across households so that each household works the same number of hours, implying that $n_{it} = N_t$. In Section 3.4 we also report results from a version of the model with sticky prices, flexible wages, and a labor supply choice at the household level.

Production A representative firm produces the final good Y_t with a linear production function that uses effective labor N_t as its only factor of production. Equilibrium therefore implies a real wage of $\frac{W_t}{P_t} = 1$ per effective unit of labor and zero profits so that $d_{it} = 0$.

Wage Phillips curve We adopt the wage setting model in Auclert et al. (2024), in which monopolistically competitive labor unions set nominal wages subject to quadratic adjustment costs to maximize the welfare of a household with aggregate consumption $C_t = \int_i c_{it} di$ and hours N_t . This leads to the wage Phillips curve

$$\pi_t = \kappa \left(v'(N_t) - \frac{1 - \tau}{C_t} \right) + \bar{\beta} \pi_{t+1}, \quad (1)$$

where π_t denotes wage (and price) inflation, κ depends on the wage adjustment cost and $\bar{\beta}$ is the average discount factor. This is the same linearized wage Phillips curve that arises in an analogous representative agent economy (e.g., Erceg et al., 2000). We choose to work directly with the linearized Phillips curve so as to focus entirely on the non-linearity of the consumption function stemming from the borrowing constraint.

Fiscal policy The government issues real bonds B_t subject to the budget constraint⁴

$$B_t - B_{t-1} = G + T - \tau Y_t + r_{t-1} B_{t-1}, \quad (2)$$

where G is a fixed level of government consumption. Outside of steady state, the government adjusts the lump-sum transfer component of the tax-and-transfer schedule according to the

⁴We present results for an economy in which the government issues nominal bonds in Section 3.4.

rule

$$T_t = T - \phi_B \bar{r}(B_{t-1} - \bar{B}) \quad (3)$$

where $\phi_B > 1$ and $\bar{r}$ and $\bar{B}$ are the steady-state levels for the interest rate and government debt. This restriction implies that transfers are set so as to ensure that debt always returns to its steady-state level. Hence, fiscal policy is passive in the language of [Leeper \(1991\)](#).

Monetary policy The central bank sets the nominal interest rate according to the Taylor rule

$$i_t = \bar{i} + \phi_\pi(\pi_t - \bar{\pi}), \quad (4)$$

where $\bar{i}$ is the steady-state nominal rate, $\bar{\pi}$ is steady-state inflation and $\phi_\pi > 1$. Hence, monetary policy is active in the language of [Leeper \(1991\)](#). A standard Fisher equation connects real and nominal rates, $1 + i_t = (1 + r_t)(1 + \pi_{t+1})$ for $t \geq 0$.

Equilibrium Given fiscal and monetary policy rules (3) and (4), and an initial distribution of households over assets, productivity and discount factors, Γ_0 , an equilibrium is a sequence of household policy functions $\{c_t(a, e, \beta), a'_t(a, e, \beta)\}_{t=0}^\infty$ and associated distributions $\{\Gamma_t(a, e, \beta)\}_{t=1}^\infty$, aggregate quantities $\{Y_t, C_t, B_t, N_t, d_t\}_{t=0}^\infty$ and prices $\{\pi_t, w_t, i_t, r_t\}_{t=0}^\infty$ such that (i) households and firms optimize; (ii) the wage Phillips curve (1) is satisfied; (iii) the government budget constraint (2) holds; (iv) the asset market clears

$$B_t = \int a'_t(a, e, \beta) d\Gamma_t(a, e, \beta) \quad \text{for } t \geq 0;$$

and (v) aggregate consumption is given by

$$C_t = \int c_t(a, e, \beta) d\Gamma_t(a, e, \beta) \quad \text{for } t \geq 0.$$

Walras' law implies the goods market clearing condition $C_t + G = Y_t$ holds.

3.2 Calibration

The calibration follows closely the parameterization of the partial equilibrium model in Section 2. Below we describe the features that are specific to the general equilibrium version. See Table A.4 in Appendix A.2 for further details.

Preferences We assume the following functional form for the disutility of labor

$$v(n) = \chi \frac{n^{1+\frac{1}{\nu}}}{1+\frac{1}{\nu}},$$

where ν is the Frisch elasticity of labor supply, which we set equal to 1. We normalize χ so that quarterly output equals one in steady state. As in Section 2, we choose the set of discount factors to match an average wealth to annual GDP ratio of 100% and an average quarterly MPC out of \$500 of 25%.

Idiosyncratic income process The stochastic process for individual efficiency units of labor e_{it} is the same as the process for household labor income in Section 2.

Wage setting We set the slope of the wage Phillips curve $\kappa = 0.01$, as in Auclert et al. (2024).

Fiscal policy We set government consumption G to 10.4% of annual GDP. The steady-state tax rate and lump sum transfer are the same as in Section 2. Hence the primary surplus as a percentage of annual GDP is $\mathcal{T}(1) - G = (0.274 - 0.15) - 0.104 = 2\%$. With a debt-to-GDP ratio of 100% this implies a steady-state interest rate of 2% p.a. We set the parameter governing the speed of repayment outside of steady-state to $\phi_B = 8$. This implies that the half-life of an increase in government debt is roughly 5 years.

Monetary policy We set steady-state inflation, $\bar{\pi} = 0$ so that the steady-state nominal rate is $\bar{i} = 2\%$. In our baseline experiments we set the Taylor rule coefficient $\phi_\pi = 1.5$.

3.3 Aggregate Transfers in General Equilibrium

Experiment As in Section 2, we study an unexpected one-time increase in the lump-sum transfer to all households at $t = 0$, which we implement by setting $T_0 = T + \Delta$. In our baseline experiment we assume that the stimulus is financed by lowering future lump-sum transfers for $t > 0$ according to the fiscal rule (3).

Computation We obtain the linear solutions using the SSJ, following Auclert et al. (2021). We compute non-linear solutions by finding the path of aggregate variables that satisfy the market clearing conditions for each size stimulus Δ .

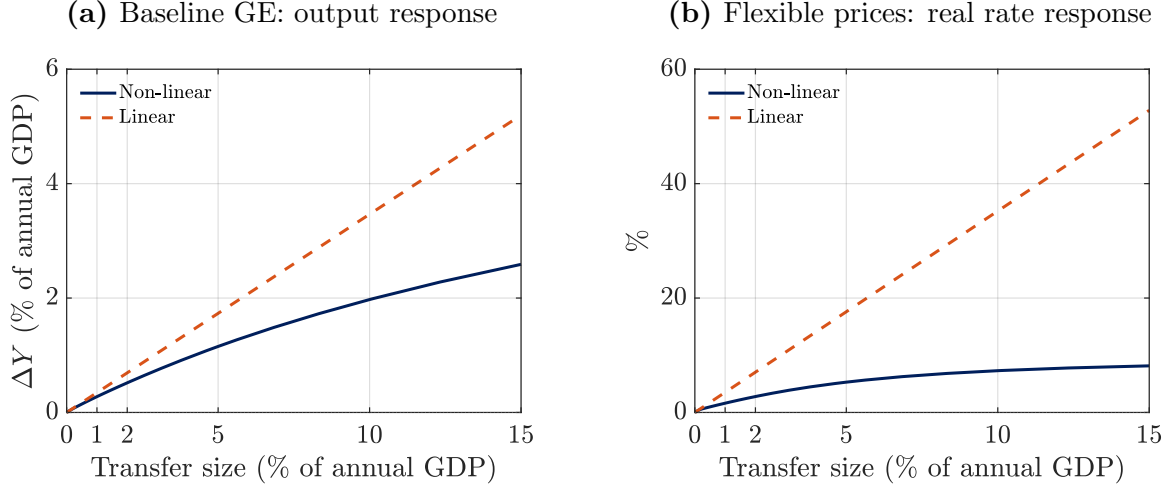


Figure 3: Aggregate effects of a one-time transfer in general equilibrium. Left panel: output response with sticky wages; Right panel: interest rate response with flexible wages

Baseline HANK results Figure 3a shows the aggregate output response for different values of Δ for the linear and non-linear solution methods. The non-linear solution is concave in the size of the shock. For a stimulus that is 2% of steady-state GDP (around \$3,500 per household), the first-quarter output response is around 0.53% of GDP according to the non-linear solution and around 0.71% according to the linear solution. The error in the linear solution is thus around 34%. For a stimulus of size 5% of steady-state GDP (around \$8,400), the first-quarter error in the linear solution is around 50%. Over longer horizons the differences between the linear and non-linear solutions are smaller, but remain economically significant. Figure F.3 in Appendix F contains the full impulse responses as well as analogous figures for first-year responses.

To understand where the errors in the linear solution stem from, it is useful to recap the mechanism for why a one-time transfer raises output in this model. Because of a failure of Ricardian equivalence, the combination of a higher transfer at $t = 0$ and lower future transfers leads households to want to increase their consumption, which leads to a rise in the natural interest rate. In the presence of nominal rigidities this leads to an inflationary output boom as long as the central bank does not fully accommodate the rise in the natural rate by raising the nominal rate.⁵

The strength of this mechanism depends on the initial upward pressure on household

⁵The precise mechanism by which output rises depends on the particular model of nominal rigidities. In this example with a monopolistic labor union, nominal wage rigidities and a fixed real wage, the union trades off the cost of wage adjustments with the utility cost of supplying labor in excess of the utility-maximizing level.

consumption, which is determined entirely by the departure from Ricardian equivalence, as reflected in the subset of households with MPCs above the PIH. Recall that in a representative agent version of this experiment there would be no effect on output for any size of stimulus. But as we saw in Figure 1, the high-MPC households are those who are on or close to the borrowing constraint and so have a consumption response that is highly non-linear in the size of the transfer. Thus the same forces that drive the concave response of consumption in the partial equilibrium experiments in Section 2 also lead to concavity in the output response in general equilibrium.

Interest rate response with flexible wages Before investigating the robustness of these findings to alternative model specifications, Figure 3b shows the interest rate response when there are no wage adjustment costs. The red dashed and blue solid lines show the linear and non-linear first-quarter response when unions continue to choose the hours on behalf of households. In this economy, the fiscal stimulus policy has no effects on employment, aggregate output or consumption. However, it does lead to movements in the real interest rate. The funded stimulus program is a mechanism to relieve the effects of borrowing constraints. Constrained households with high MPCs increase their consumption in response to the shock. Since output is unaffected, the interest rate must rise to induce higher wealth households to cut their consumption. Once again, because the strength of these effects is tied to the shape of the consumption function for constrained households, the effect is weaker for larger values of Δ .

Figure F.1 in Appendix F shows the interest rate response in an analogous version of the model without a labor union, in which individual households can choose labor supply freely. In this model there are some small effects on output because of different effects on labor supply incentives for households with different productivity and asset levels. But since the extent of non-linearity is ultimately tied to the failure of Ricardian equivalence, the aggregate effect looks similar to the version with a union.

3.4 Other Versions of the HANK Model

The baseline general equilibrium model in the previous section features sticky wages, flexible prices, real government debt and an active monetary, passive fiscal policy configuration. In this section we investigate whether the extent of non-linearity in the effects of fiscal stimulus is affected by changes in these model features. The findings are summarized in Figure F.2.

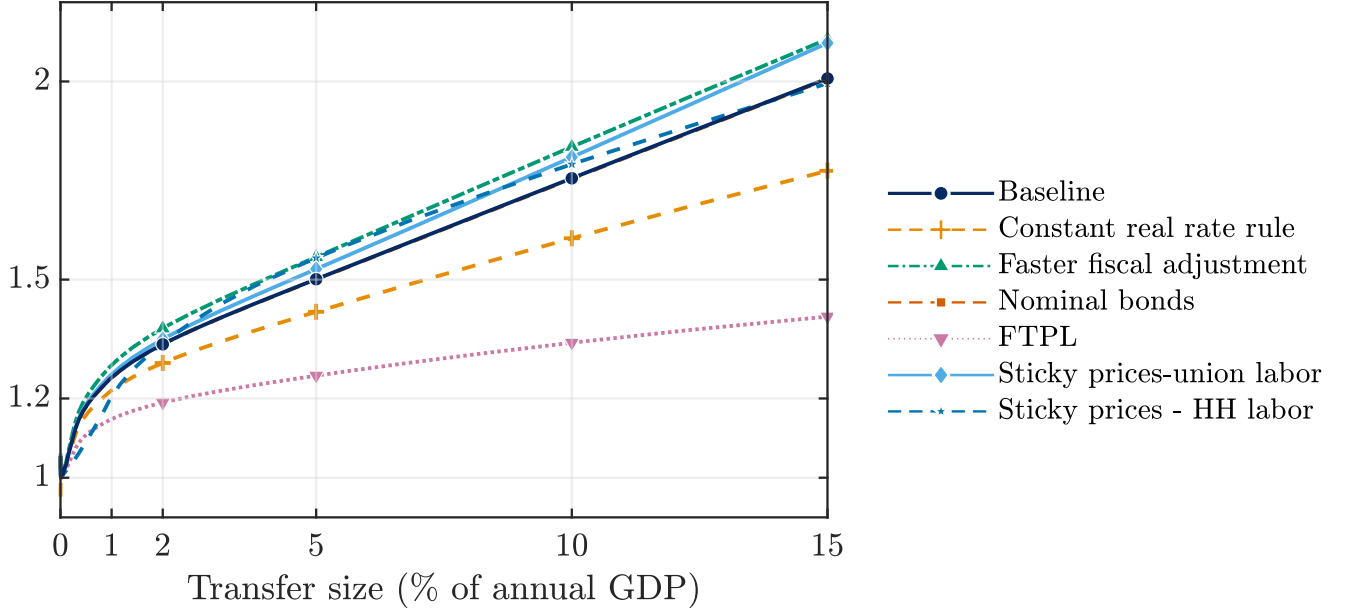


Figure 4: Sensitivity in HANK

Notes: The figure presents the ratio of the linear to the non-linear output response for different model calibrations.

Faster fiscal adjustment. The green dash-dot line labeled “Faster fiscal adjustment” in Figure F.2 shows that the non-linearity is more pronounced when the fiscal authority stabilizes debt more rapidly with a fiscal rule that sets $\phi_B = 16$ compared with $\phi_B = 8$ in the baseline. Since future transfers are lowered sooner it is only the most constrained households, those whose departure from Ricardian equivalence is most severe, whose consumption is affected. And as we have seen, these households have the most non-linear consumption functions.

Fixed real rate The orange dashed line labeled “Constant real rate rule” in Figure F.2 shows results from an economy where the central bank keeps the real rate constant. This economy has slightly less non-linearity than the baseline economy.

Sticky prices and flex wages An alternative model of nominal rigidities is one with sticky prices and flexible wages. We model sticky prices by introducing an intermediate-goods sector with a CES aggregator, monopolistic competition and quadratic price adjustment costs as in Rotemberg (1982). This version of the model features a standard New Keynesian Phillips curve of the form

$$\pi_t = \kappa \left(\frac{W_t}{P_t} - 1 \right) + \beta \pi_{t+1}$$

Figure F.2 displays results for two versions, one in which households freely choose how much labor to supply (light blue solid line), and one in which the union chooses a common number of hours for all households (light blue dashed line).⁶ Both versions exhibit a degree of non-linearity similar to that in the baseline model.

Unfunded fiscal stimulus We also consider a version of the model in which future inflation, rather than future primary surpluses, pays for the stimulus, as in the Fiscal Theory of the Price Level. To implement this version we make two changes to the baseline model.

First, we assume that the government issues nominal, rather than real debt. This on its own has a negligible effect on the extent of non-linearity, as illustrated by the purple dashed line labeled “Nominal bonds” in Figure F.2. The only effect of this change is that the unexpected inflation in the first period following the stimulus lowers the real value of debt that must be repaid with lower future surpluses.

Second, we alter the fiscal and monetary rules so that $\phi_B = \phi_\pi = 0$. This has the effect of pegging the nominal rate at $i_t = \bar{i}$ and holding real transfers fixed at their steady-state level $T_t = T$ for $t > 0$. Hence, in the language of [Leeper \(1991\)](#), fiscal policy is active and monetary policy is passive.

The ratio of the linear to non-linear first-quarter output response is shown with the pink dotted line labeled “FTPL” in Figure F.2. The figure shows that for an unfunded stimulus the extent of non-linearity is much less severe than for the other versions of the model.⁷ The reason is that a large part of the output effect of unfunded stimulus does not rely on a failure of Ricardian equivalence or households having MPCs above the PIH. Even in a representative agent model, an unfunded stimulus of this type has an expansionary effect on output. The upward pressure on consumption arises because households seek to exchange nominal government debt for real consumption, as explained in [Cochrane \(2023\)](#). Redistribution towards high-MPC households plays a role as well, as explained by [Kaplan, Nikolakoudis and Violante \(2024\)](#), but it is a small part of the overall effect on output.

⁶When households choose labor supply, they set n_{it} so that $\frac{1}{c_{it}}e_{it}(1-\tau)\frac{W_t}{P_t} = v'(n_{it})$. When the labor union chooses labor supply they set N_t so that $\frac{1}{C_t}\frac{W_t}{P_t}(1-\tau) = v'(n_t)$.

⁷Figure F.4 in Appendix F reports the aggregate output effect of the unfunded stimulus, which is larger than for the funded stimulus.

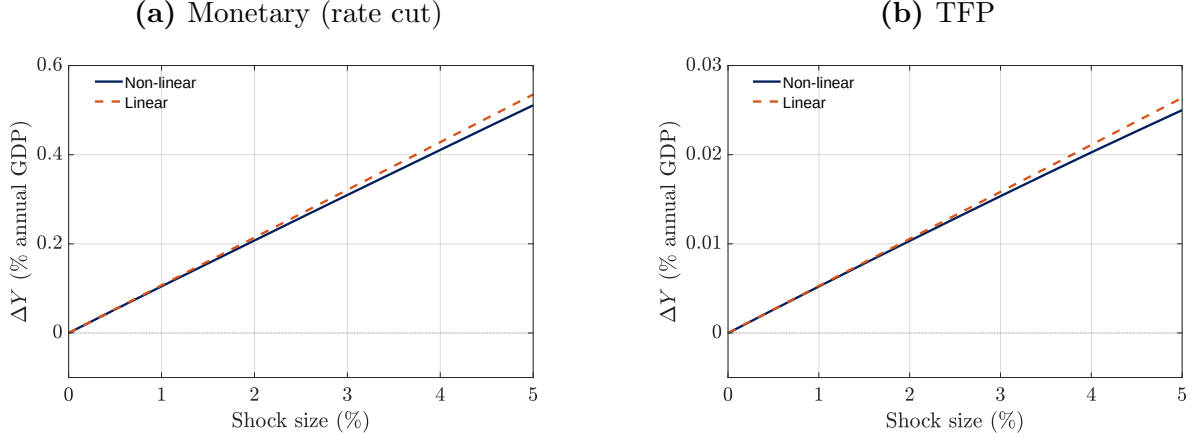


Figure 5: Non-linearities for Monetary Policy and TFP.

Note: the figures present the output response as a function of monetary shocks (pp annualized) and TFP shocks. The monetary shock is Δ_0 such that $i_t = \bar{i} + \phi_\pi(\pi_t - \bar{\pi}) - \varepsilon_i$. The productivity shock is given by Z_0 .

4 Other Shocks

We have focused on a particular policy experiment—a one-time lump-sum fiscal transfer—in which a failure of Ricardian equivalence is crucial for the aggregate effects. In this section we discuss other types of shocks in which Ricardian equivalence plays less of a role, and show that for these shocks, the non-linearities are less severe.

Monetary shocks Figure 5a compares the first-quarter output response to a one-quarter cut in the nominal rate of different sizes, using linear and non-linear solution methods. The two solution methods yield almost identical responses, consistent with simulations in Auclert et al. (2021). This should not be surprising, since we know from Werning (2015) and Kaplan et al. (2018) that heterogeneity has only a small impact on the aggregate effects of monetary policy shocks. Rather, the nature of MPCs affects the transmission mechanism and decomposition between direct and indirect effects.

Productivity shocks Figure 5b shows the first-quarter response to an aggregate productivity shock. We model the shock by modifying the production function to be $Y_t = e^{Z_t} N_t$ and consider a one-time shock to Z_0 . The figure compares the linear and non-linear solution methods for different shock sizes. Again, the two solution methods yield almost identical responses, which are very close to those in an analogous representative agent economy. The

reason is that failures of Ricardian equivalence, and hence the size and nature of MPCs, play a minimal role in the transmission of productivity shocks.

Additional shocks In Appendix Figure F.5 we present analogous results for shocks to the household discount factor and to the labor tax rates. In particular, labor tax cuts transfer fewer resources to low-productivity, borrowing-constrained households and therefore generate smaller non-linearities than lump-sum transfers.

5 Conclusions

Heterogeneous agent models are useful for studying the effects of fiscal stimulus because the interaction between borrowing constraints and incomplete markets for idiosyncratic risk leads to a failure of Ricardian equivalence. However, the resulting MPCs are highly non-linear in household wealth. As a result, the aggregate effects of fiscal transfers to households with high MPCs decline rapidly with their size. Using first-order methods in this class of models effectively treats constrained households as if they had an MPC of 1 regardless of the size of the transfer and overstates their true response. For some shocks and policy experiments, this approximation is accurate. But for local linear methods such as SSJ to yield reliable quantitative results when studying fiscal stimulus—particularly redistributive stimulus targeted to high-MPC households—then “small” must be very small indeed.

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Online Appendix: Income Process Estimation

How Small is Small? Non-linearities in Heterogeneous Agent Models

Javier Bianchi and Greg Kaplan

A Calibration

A.1 Income Process

This appendix describes the estimation of the income process used in our quantitative analysis. We follow closely the approach of [Kaplan and Violante \(2022\)](#) (hereafter KV).

Panel Study of Income Dynamics

We use data from the Panel Study of Income Dynamics (PSID) on total annual household labor income for households with heads aged 25 to 65 from 1968 to 2008, following KV. We drop households with annual labor income less than \$7,250 in 2016 dollars, which corresponds to 1,000 hours per year at \$7.25 per hour (part-time employment at the minimum wage). We remove age and year effects in a first stage by regressing household labor income on a full set of year and age dummies and we construct the empirical counterparts to $m_{2,d}$ using the residuals from this regression.

Let $\log y_t^{ann}$ be annual labor income in year t , and let annual income growth at lag d be

$$\Delta_d \log y_t^{ann} = \begin{cases} \log y_t^{ann} & \text{if } d = 0, \\ \log y_{t+d}^{ann} - \log y_t^{ann} & \text{if } d > 0. \end{cases}$$

Define cross-sectional moments of annual income growth of order j at lag d as

$$m_{j,d} = \mathbb{E} \left[(\Delta_d \log y_t^{ann})^j \right]$$

and the kurtosis of income growth at lag d as

$$\kappa_d = \frac{m_{4,d}}{(m_{2,d})^2}.$$

Table [A.1](#) reports the empirical estimates for the cross-sectional moments that we use in estimation, which coincide with those in KV.

Table A.1: Empirical moments of annual income growth at different lags.
Source: PSID 1968–2008.

Lag (d)	$m_{2,d}$	$m_{4,d}$	κ_d
0	0.504	0.930	3.65
1	0.142	0.220	10.90
2	0.207	0.369	8.57
3	0.235	0.410	7.42
4	0.280	0.544	6.96
5	0.295	0.557	6.39

Estimated Income Processes

We model the discrete-time quarterly income process, as in KV:

$$\log y_s = \begin{cases} z_t + \varepsilon_t & \text{with probability } \lambda_\varepsilon, \\ z_t & \text{with probability } 1 - \lambda_\varepsilon, \end{cases} \quad \varepsilon_t \sim \mathcal{N}\left(-\frac{\sigma_\varepsilon^2}{2}, \sigma_\varepsilon^2\right),$$

$$z_t = \begin{cases} \phi_z z_{t-1} + \eta_t & \text{with probability } \lambda_\eta, \\ \phi_z z_{t-1} & \text{with probability } 1 - \lambda_\eta. \end{cases} \quad \eta_t \sim \mathcal{N}\left(-\frac{\sigma_\eta^2}{2}, \sigma_\eta^2\right),$$

We define annual income y^{ann} as the sum of the four quarterly incomes within the year. Based on this definition, we construct the model counterparts of all the empirical moments in Table A.1.

With $\lambda_\varepsilon = \lambda_\eta = 1/4$ fixed—so that each type of shock arrives on average once per year—we have three free parameters $(\phi_z, \sigma_\eta^2, \sigma_\varepsilon^2)$. We estimate them by minimizing the sum of squared deviations between model and data second moments $(m_{2,0}, m_{2,1}, m_{2,5})$.

We discretize the shock process using $n_z = 11$ Rouwenhorst states for the persistent component and $n_\varepsilon = 5$ points for the transitory component, for a total of 55 income states.

Table A.2 reports the estimated parameters. The persistent component has quarterly autocorrelation $\phi_z = 0.9878$ and innovation variance $\sigma_\eta^2 = 0.0439$; the transitory shock has variance $\sigma_\varepsilon^2 = 0.6376$, implying large but infrequent income fluctuations.

Specifically, the transitory mixture distribution $(1 - \lambda_\varepsilon) \delta_0 + \lambda_\varepsilon \mathcal{N}(-\sigma_\varepsilon^2/2, \sigma_\varepsilon^2)$ is approximated with $n_\varepsilon = 5$ points by solving for the grid and probability weights that match its second and fourth central moments. The resulting grid spans $\log y^\varepsilon \in \{-2.85, -1.42, 0, 1.42, 2.85\}$

with probability weights $\{\approx 0, 0.042, 0.917, 0.042, \approx 0\}$, placing 91.7% of the mass at the no-shock point, consistent with the 75% no-shock probability from $\lambda_\varepsilon = 0.25$ plus the mass contributed by the discretized normal at zero.

Table A.2: Parameter estimates for the quarterly income process.

Parameter	Description	Symbol	Value
<i>A. Structural parameters</i>			
AR(1) persistence	Persistent component	ϕ_z	0.9878
Persistent shock variance	Innovation variance	σ_η^2	0.0439
Transitory shock variance	Transitory variance	σ_ε^2	0.6376
Persistent arrival rate	Quarterly probability	λ_η	0.250
Transitory arrival rate	Quarterly probability	λ_ε	0.250
<i>B. Discretization</i>			
Persistent states	Rouwenhorst grid	n_z	11
Transitory states	Mixture approximation	n_ε	5
Total income states		$n_z \times n_\varepsilon$	55
<i>C. Targeted moments</i>			
Variance, lag 0		$m_{2,0}$	0.504
Variance, lag 1		$m_{2,1}$	0.142
Variance, lag 5		$m_{2,5}$	0.295

A.2 Further Details on Calibration

Table A.3: Calibration: parameter values and steady-state moments across PE scenarios

Scenario	Parameters		Moments	
	$\bar{\beta}$	σ_{β}	Avg. MPC	A/Y
Baseline	0.972	0.010	0.25	1.00
No β heterog., $A/Y = 1$	0.991	0	0.09	1.00
No β heterog., $A/Y = 4$	0.993	0	0.05	4.00
Stochastic β	0.993	0.066	0.25	1.00
Mortality	0.978	0.011	0.25	1.00
Lower transfers	0.966	0.013	0.25	1.00
Natural debt limit	0.961	0.016	0.25	1.00
Endogenous labor supply	0.956	0.018	0.32	1.00
ARS income process	0.976	0.006	0.25	1.00

Note: Avg. MPC is the steady-state average marginal propensity to consume. The versions of the model are as follows: *No β heterog. ($A/Y=1$)* and *No β heterog. ($A/Y=4$)* are versions without discount-factor heterogeneity, calibrated to match aggregate wealth-to-(annualized) income ratios of 1 and 4, respectively. *Stochastic β* : discount factors follow an idiosyncratic stochastic process AR(1) with autocorrelation $\rho_{\beta} = 0.990$. *Finite lives*: a life-cycle version with overlapping generations and a annual mortality probability of 2%. *Lower transfers*: steady-state lump-sum transfers are set to half of the baseline value. *Natural debt limit*: households face a 0.1% probability of transitioning to a zero-income state and setting lump-sum transfers to zero. *Endogenous labor supply*: households choose hours worked endogenously with isoelastic disutility from labor and unitary Frisch elasticity. *ARS*: replaces the baseline idiosyncratic earnings process with the discretized process from Auclert et al. (2024).

Table A.4: Summary of parameters for general equilibrium model

Parameter	Value	Description
$\bar{B}$	4	Steady-state government debt
$\bar{r}$	0.02	Steady-state annual real interest rate
$\bar{\pi}$	0	Steady-state inflation
κ	0.01	Phillips curve slope
ν	1	Frisch elasticity of labor supply
ϕ_{π}	1.5	Taylor coefficient
ϕ_B	8	Fiscal rule
G	0.104	Government spending
T	0.15	Government Transfers
χ	1	Labor disutility

B Analytical Characterization of MPCs

This appendix derives analytical first-order and higher-order derivatives of the individual consumption function.

Let cash-on-hand be

$$x \equiv Ra + y.$$

The budget constraint is

$$c + a' = x, \quad a' \geq \underline{a}.$$

Write the consumption policy as $c = c(x, y)$ and define the MPC with respect to cash-on-hand by⁸

$$m(x, y) \equiv c_x(x, y).$$

Next-period cash-on-hand is

$$x' = R[x - c(x, y)] + y'.$$

Let $\bar{x}(y)$ denote a borrowing-constraint boundary point: locally, the constraint binds for $x < \bar{x}(y)$ and is slack for $x > \bar{x}(y)$. Define the one-sided MPCs at the boundary by

$$m^-(\bar{x}(y), y) \equiv \lim_{\varepsilon \downarrow 0} m(\bar{x}(y) - \varepsilon, y), \quad m^+(\bar{x}(y), y) \equiv \lim_{\varepsilon \downarrow 0} m(\bar{x}(y) + \varepsilon, y).$$

Let $V(a, y)$ denote the beginning-of-period value function, with asset state a . Given cash-on-hand x , the household solves

$$\max_{a' \geq \underline{a}} \{u(x - a') + \beta \mathbb{E}[V(a', y') \mid y]\}.$$

The consumption policy is

$$c(x, y) = x - a'(x, y).$$

For the slack branch, use the notation

$$c \equiv c(x, y), \quad c' \equiv c(x', y'), \quad m \equiv m(x, y), \quad m' \equiv m(x', y'),$$

and

$$(m_x)' \equiv m_x(x', y'), \quad (m_{xx})' \equiv m_{xx}(x', y').$$

⁸The consumption policy varies across households with different β . We do not index them to simplify notation.

Define

$$\begin{aligned}\Lambda &\equiv \beta R^2 \mathbb{E} [u''(c')m' \mid y], \\ \Psi &\equiv \beta R \mathbb{E} [u'''(c')(m')^2 + u''(c')(m_x)' \mid y], \\ \Theta &\equiv \beta R \mathbb{E} [u''''(c')(m')^3 + 3u'''(c')m'(m_x)' + u''(c')(m_{xx})' \mid y].\end{aligned}$$

Constrained branch. First consider the constrained branch. If the borrowing constraint binds, then

$$a'(x, y) = \underline{a}.$$

Therefore

$$c(x, y) = x - \underline{a}.$$

Differentiating with respect to cash-on-hand gives

$$m(x, y) = c_x(x, y) = 1, \quad m_x(x, y) = c_{xx}(x, y) = 0, \quad m_{xx}(x, y) = c_{xxx}(x, y) = 0. \quad (\text{B.1})$$

Thus

$$m^-(\bar{x}(y), y) = 1.$$

Slack branch: first derivative. Now consider the slack branch, where $a'(x, y) > \underline{a}$. The first-order condition is

$$u'(c(x, y)) = \beta \mathbb{E} [V_a(a'(x, y), y') \mid y]. \quad (\text{B.2})$$

At smooth continuation points, differentiating (B.2) with respect to x gives

$$u''(c)m = \beta \mathbb{E} [V_{aa}(a'(x, y), y') \mid y] a'_x.$$

Since

$$a'_x = 1 - c_x(x, y) = 1 - m(x, y),$$

we obtain

$$u''(c)m = \tilde{\Lambda}(1 - m), \quad (\text{B.3})$$

where

$$\tilde{\Lambda} \equiv \beta \mathbb{E} [V_{aa}(a'(x, y), y') \mid y].$$

If the continuation value is strictly concave at the smooth continuation points reached from the slack branch, then

$$\tilde{\Lambda} < 0.$$

Since $u''(c) < 0$, equation (B.3) implies

$$m(x, y) = \frac{\tilde{\Lambda}}{u''(c) + \tilde{\Lambda}}, \quad 0 < m(x, y) < 1. \quad (\text{B.4})$$

Thus the slack-side MPC is strictly below the constrained-side MPC.

The expression above can be written in terms of the Euler equation. At a smooth continuation point, the envelope condition gives

$$V_a(a', y') = Ru'(c(x', y')), \quad x' = Ra' + y'. \quad (\text{B.5})$$

Differentiating (B.5) with respect to a' gives

$$V_{aa}(a', y') = R^2 u''(c') m'.$$

Therefore

$$\tilde{\Lambda} = \beta R^2 \mathbb{E}[u''(c') m' \mid y] \equiv \Lambda.$$

Substituting this into (B.4) gives

$$m(x, y) = \frac{\Lambda}{u''(c) + \Lambda}. \quad (\text{B.6})$$

Slack branch: second derivative. The Euler equation on the slack branch is

$$u'(c) = \beta R \mathbb{E}[u'(c') \mid y]. \quad (\text{B.7})$$

Define

$$F(z, y') \equiv u'(c(z, y')).$$

Then, evaluated at $z = x'$,

$$F_z = u''(c') m', \quad (\text{B.8})$$

$$F_{zz} = u'''(c') (m')^2 + u''(c') (m_x)', \quad (\text{B.9})$$

$$F_{zzz} = u''''(c') (m')^3 + 3u'''(c') m' (m_x)' + u''(c') (m_{xx})'. \quad (\text{B.10})$$

Also,

$$x'_x = R(1 - m), \quad x'_{xx} = -Rm_x, \quad x'_{xxx} = -Rm_{xx}. \quad (\text{B.11})$$

Differentiating (B.7) twice with respect to x gives

$$u'''(c)m^2 + u''(c)m_x = \beta R \mathbb{E} [F_{zz}(x'_x)^2 + F_z x'_{xx} \mid y].$$

Using (B.8), (B.9), and (B.11), this becomes

$$u'''(c)m^2 + u''(c)m_x = [R(1 - m)]^2 \Psi - \Lambda m_x.$$

Rearranging yields

$$m_x(x, y) = \frac{[R(1 - m)]^2 \Psi - u'''(c)m^2}{u''(c) + \Lambda}. \quad (\text{B.12})$$

Slack branch: third derivative. Differentiating (B.7) three times with respect to x gives

$$u''''(c)m^3 + 3u'''(c)mm_x + u''(c)m_{xx} = \beta R \mathbb{E} [F_{zzz}(x'_x)^3 + 3F_{zz}x'_x x'_{xx} + F_z x'_{xxx} \mid y].$$

Using (B.8), (B.9), (B.10), and (B.11), the right-hand side is

$$[R(1 - m)]^3 \Theta - 3R^2(1 - m)m_x \Psi - \Lambda m_{xx}.$$

Therefore

$$u''''(c)m^3 + 3u'''(c)mm_x + u''(c)m_{xx} = [R(1 - m)]^3 \Theta - 3R^2(1 - m)m_x \Psi - \Lambda m_{xx}.$$

Rearranging gives

$$m_{xx}(x, y) = \frac{[R(1 - m)]^3 \Theta - 3R^2(1 - m)m_x \Psi - u''''(c)m^3 - 3u'''(c)mm_x}{u''(c) + \Lambda}. \quad (\text{B.13})$$

Summary. Combining the constrained-branch derivatives in (B.1) with the slack-branch formulas in (B.6), (B.12), and (B.13), we obtain

$$\begin{aligned}
m(x, y) &= \begin{cases} 1, & x < \bar{x}(y), \\ \frac{\Lambda}{u''(c) + \Lambda}, & x > \bar{x}(y), \end{cases} \\
m_x(x, y) &= \begin{cases} 0, & x < \bar{x}(y), \\ \frac{[R(1 - m)]^2 \Psi - u'''(c)m^2}{u''(c) + \Lambda}, & x > \bar{x}(y), \end{cases} \\
m_{xx}(x, y) &= \begin{cases} 0, & x < \bar{x}(y), \\ \frac{[R(1 - m)]^3 \Theta - 3R^2(1 - m)m_x \Psi - u''''(c)m^3 - 3u'''(c)mm_x}{u''(c) + \Lambda}, & x > \bar{x}(y). \end{cases}
\end{aligned}$$

In particular,

$$m^-(\bar{x}(y), y) = 1, \quad 0 < m^+(\bar{x}(y), y) < 1,$$

so the MPC is discontinuous at the point where the borrowing constraint becomes slack.

C Numerical Algorithm

C.1 Partial Equilibrium Algorithm

In partial equilibrium, the path for prices is exogenous, so the problem reduces to solving the household block and aggregating over the induced distribution. The algorithm is as follows.

1. Discretize the idiosyncratic income on finite Markov chains (see Appendix A) and choose an asset grid $\mathcal{A}$.
2. For a given transfer path, solve the household problem using the endogenous-grid method (Carroll, 2006), interpolating linearly for assets not in the grid.
3. Given the policy rules, propagate the cross-sectional distribution forward until it converges to the ergodic distribution $\Gamma(a, y, \beta)$.
4. Compute the linear response by finite differences. We solve the full household transition under a small transfer ε at $t = 0$. At $t = 0$,

$$s_0 = \frac{C_0^{ml}(+\varepsilon) - C_0^{ml}(-\varepsilon)}{2\varepsilon},$$

where

$$C_0^{ml}(\varepsilon) \equiv \int c_i(a + \varepsilon, y_i; i) d\Gamma_0(a, i),$$

and $c_i(a + \varepsilon, y)$ is the policy function for an individual with assets a , receiving transfer ε and discount factor β_i and Γ_t is evaluated along the transition path induced by the transfer. We use $\varepsilon = 1.0 \times 10^{-5}$. Automatic differentiation and central difference yield differences in the order of 10^{-7} . For ε that is not in the grid for $c(\cdot)$, we interpolate linearly.

Then, using the first-order derivative s_0 compute the linear impulse response to a transfer of size Δ as

$$\Delta C_0^{lin} = \Delta \cdot s_0.$$

For $t \geq 1$

$$C_t^{ml}(\varepsilon) \equiv \int c_i(a, y) d\Gamma_t(a, i),$$

5. Compute the non-linear response by evaluating the individual consumption policies at the transfer and aggregating using the distribution, which is propagated forward. That is,

$$C_t^m(\Delta) \equiv \int c(a, y_i; \Delta) d\Gamma_t(a, i; \Delta),$$

For the asset grid we use an exponential grid with $n_a = 200$ points on $[0, \bar{a}]$, where $\bar{a} = 4,000$ (approximately 47 times annual per-capita income in the model).

D General Equilibrium

Let the unknown transition path be

$$X \equiv \{C_t, N_t, Y_t, \pi_t, i_t, r_t, B_t\}_{t=0}^{T-1}.$$

We compute the first-order transition using the sequence-space Jacobian method of [Auclert et al. \(2021\)](#).

To solve the household problem nonlinearly, we first conjecture a vector of sequences X , and then solve backward for policy functions using EGM and then propagate the distribution forward. This yields household asset demand $A_t(X; s)$ at each date. We then evaluate the residuals from the equilibrium conditions and iterate with a modified Newton method.

Specifically, we conjecture $\{Y_t, \pi_t\}_{t=0}^{T-1}$ and then obtain $\{C_t, N_t, i_t, B_t, r_t\}_{t=0}^{T-1}$ using $Y_t = N_t = C_t + G$ and

$$\begin{aligned} i_t &= \bar{i} + \phi_\pi(\pi_t - \bar{\pi}), \\ 1 + r_t &= \frac{1 + i_t}{1 + \pi_{t+1}}, \\ B_t &= (1 + r_{t-1})B_{t-1} + G_t + T_t - \tau_t Y_t. \end{aligned}$$

The nonlinear equilibrium residuals for all $t \geq 0$ are

$$\begin{aligned} F_t^A(X; s) &= A_t(X; s) - B_t(X; s), \\ F_t^\pi(X; s) &= \kappa \left(v'(N_t) - \frac{1 - \tau}{C_t} \right) + \bar{\beta} \pi_{t+1} - \pi_t, \end{aligned}$$

Given the conjectured X , we solve backward for policy functions using EGM, propagate the distribution forward, compute asset demands and evaluate residuals. We update the conjectured sequences by a modified Newton method.

E Additional Figures: Partial Equilibrium

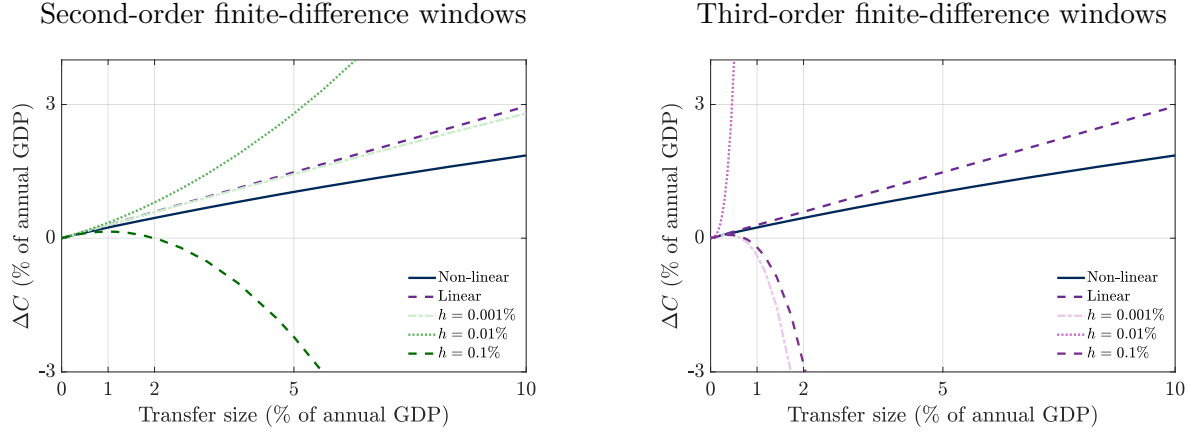
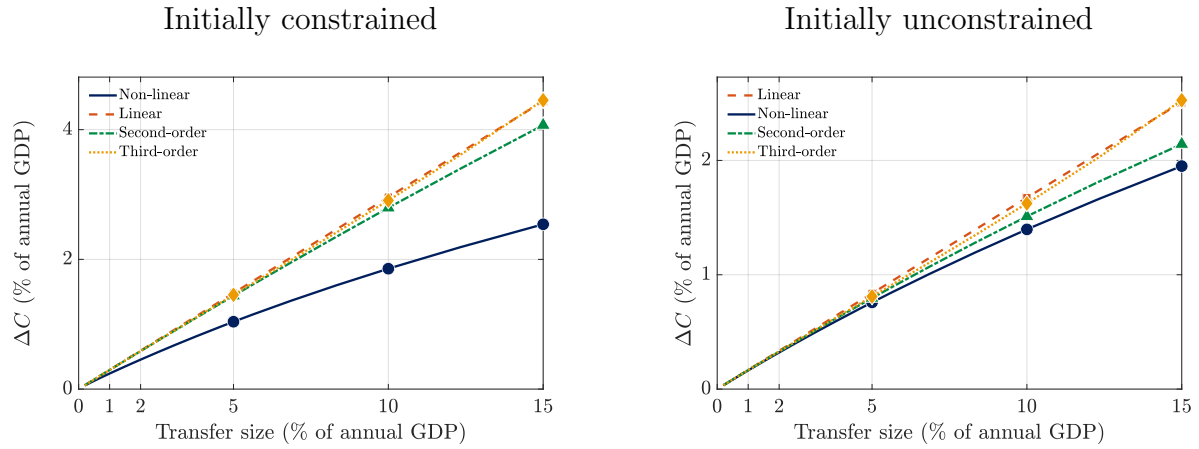


Figure E.1: Higher-order finite-difference approximations under different Richardson windows. Each panel compares the non-linear PE response, the linear response, and local higher-order approximations using three different windows: $h = 0.001\%$, 0.01% , and 0.1% of annual GDP.

Figure E.2: Constrained vs. unconstrained households



Note: Consumption response within each group, as a percentage of annual GDP per capita.

Figure E.3: Consumption function for low income

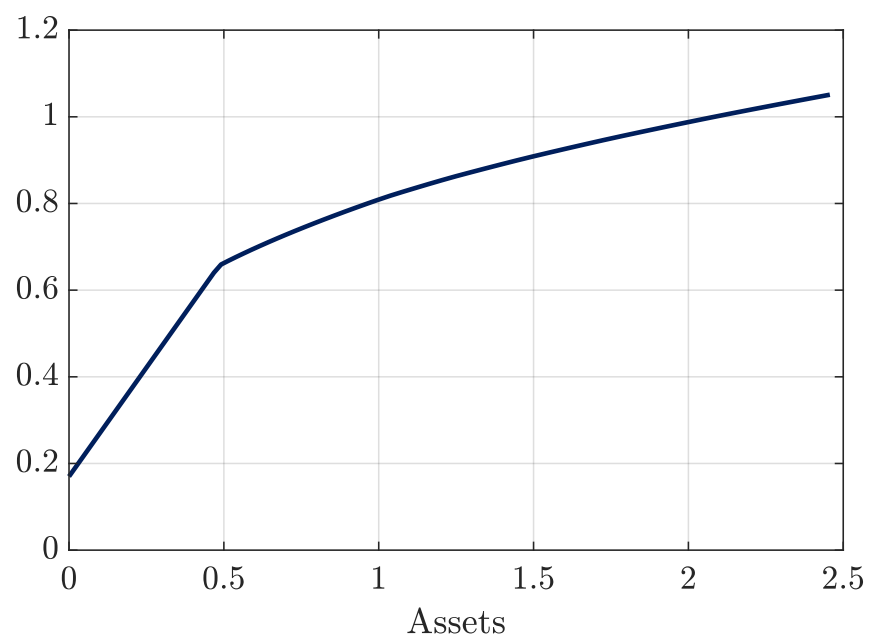


Figure E.4: Change in the Share of Constrained Households at $t = 0$

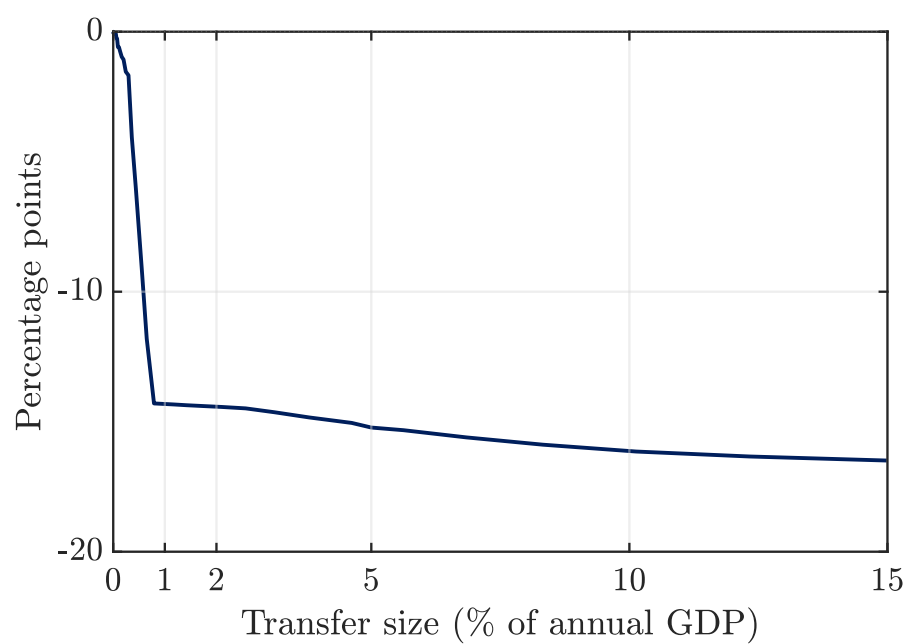
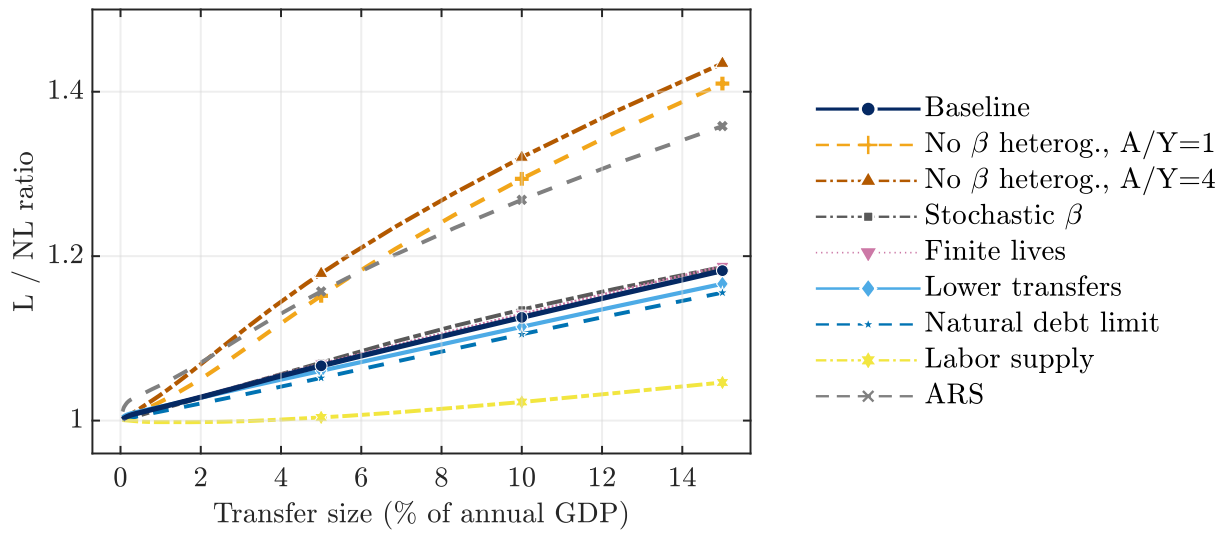


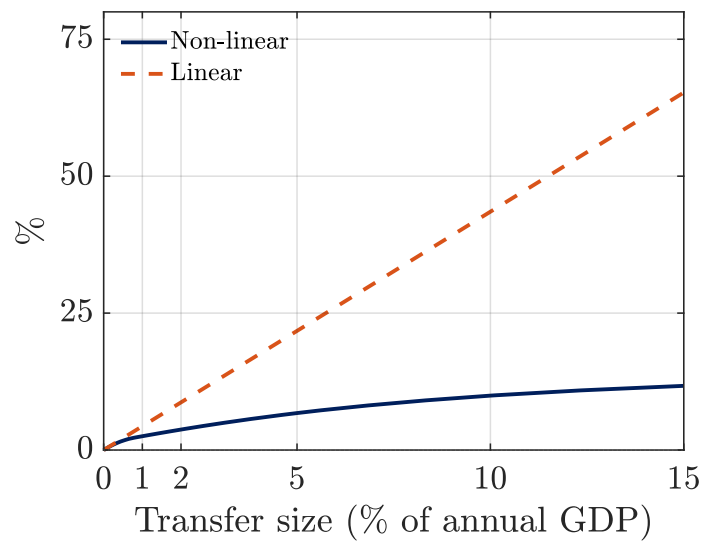
Figure E.5: First-year response: Linear to non-linear aggregate consumption response



Note: The figure presents the ratio of the 12-month linear to non-linear output response in the partial equilibrium model.

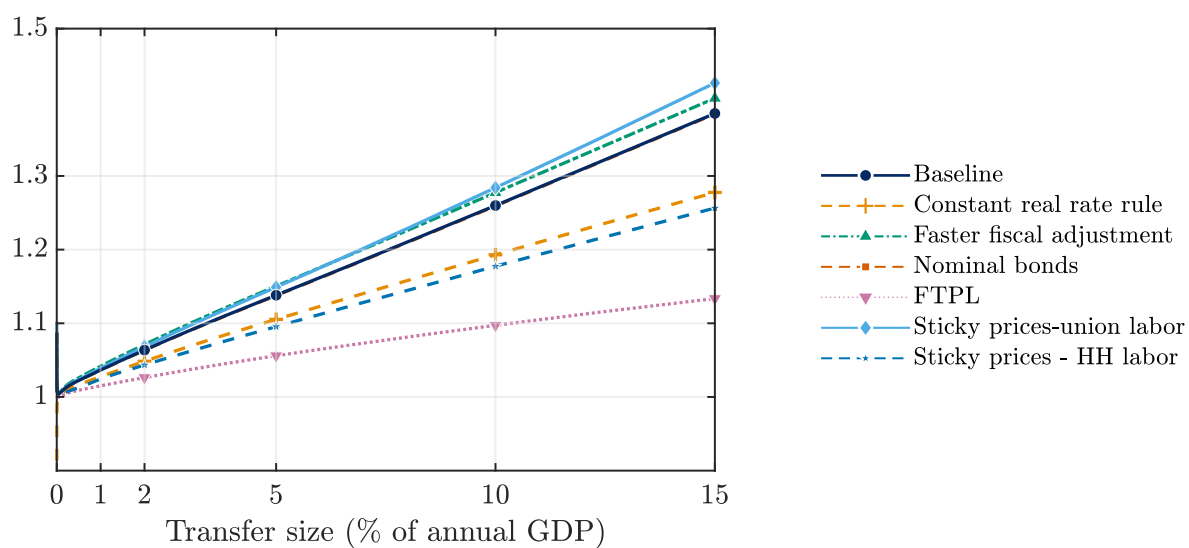
F Additional Figures: General Equilibrium

Figure F.1: Real Interest Rate Response under Flexible Prices



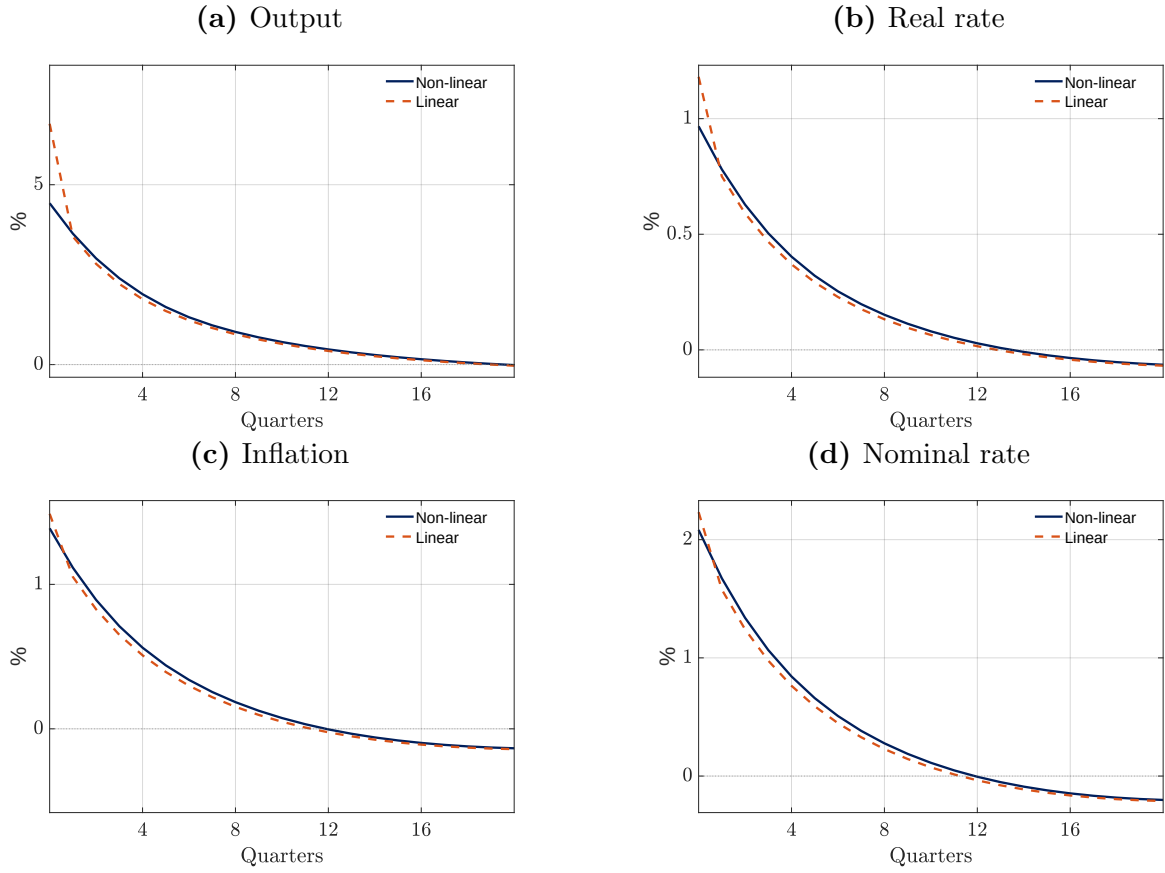
Note: real interest rate in flex-price economy in which households choose labor.

Figure F.2: Sensitivity in HANK: Annual Responses



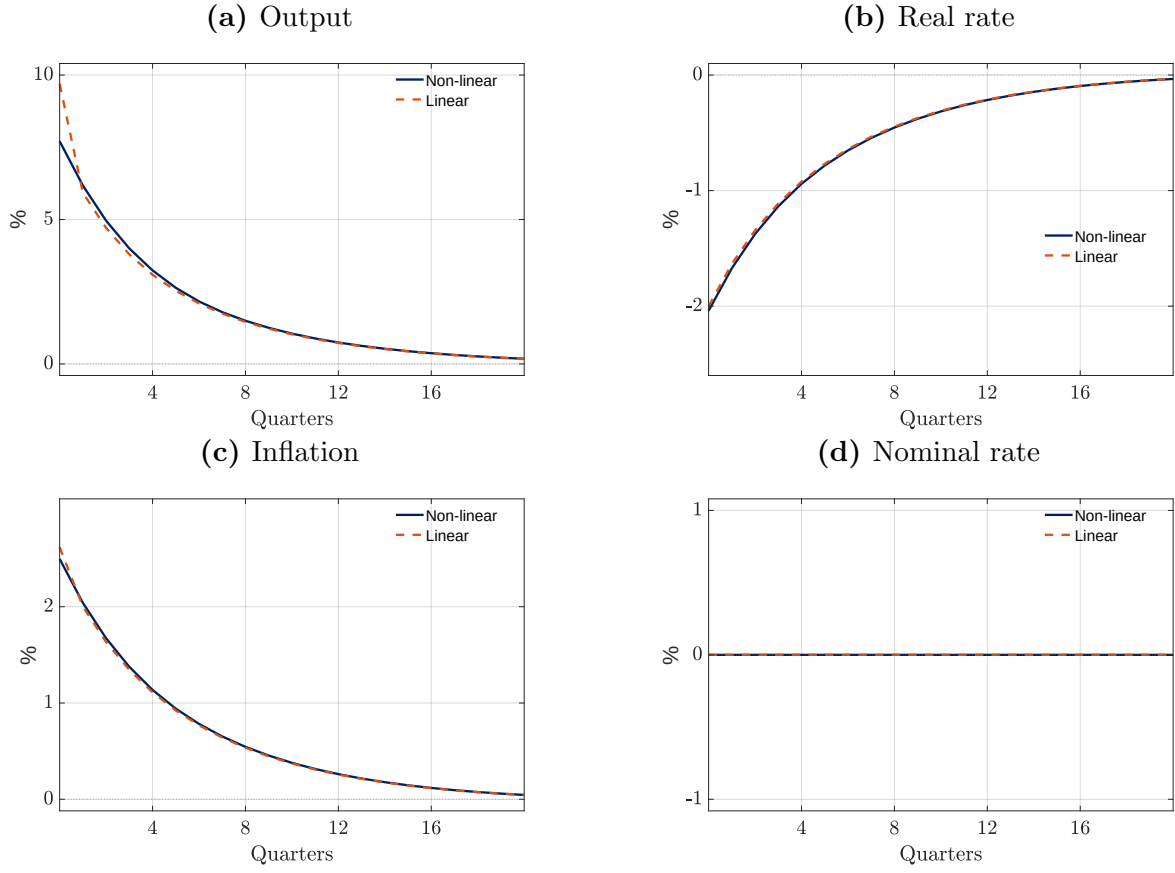
Note: The figure presents the ratio of the 12-month linear to non-linear output response in the general equilibrium model.

Figure F.3: Impulse responses baseline HANK



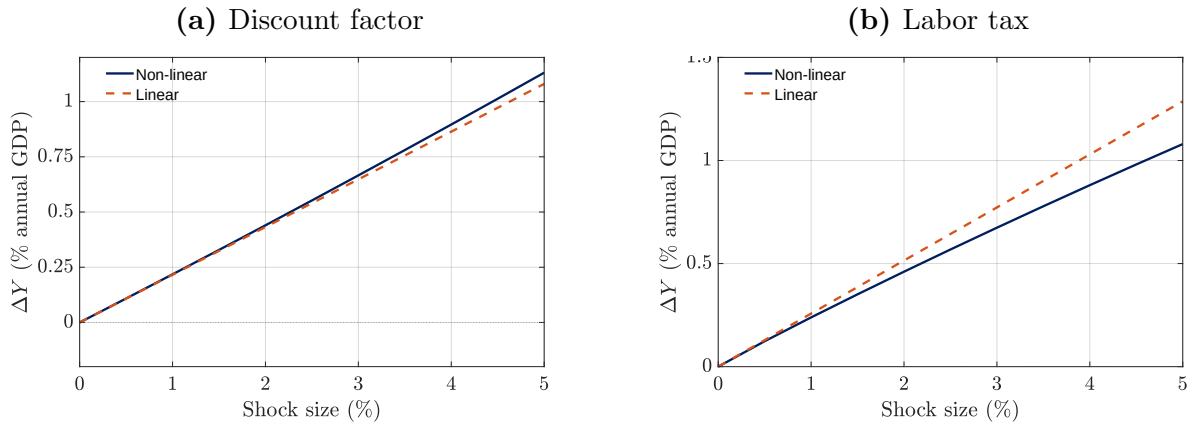
Note: output, real rate, inflation, and nominal rate for a 5% transfer. Output is in % deviations from steady state; rates are in annualized percentage points.

Figure F.4: Impulse responses: FTPL (unfunded fiscal stimulus)



Note: output, real rate, inflation, and nominal rate for a 5% transfer. Output is in % deviations from steady state; rates are in annualized percentage points.

Figure F.5: Non-linearities in response to other shocks.



Note: the discount factor shock is $\beta_{it} = \beta_{it}(1 - \varepsilon_\beta)$ and the labor tax shock is $\tau_0 = \tau(1 - \varepsilon_\tau)$, where ε_β and ε_τ represent shock size.